

MKT3434 - Introduction to Machine Learning

Assignment 1: GUI Enhancements for Data Handling and Model Completion

Introduction:

This assignment enhanced the GUI by adding support for multiple loss functions, SVM for classification and regression, and efficient missing data handling, optimizing it for better machine learning experimentation and usability.

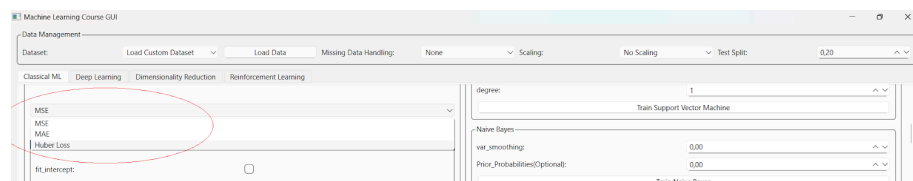
GitHub Repository: https://github.com/chaimaanairi/MKT3434_2025/tree/23501057

1. Loss Function Selection for Model Training

A new feature allows users to select loss functions for regression and classification tasks:

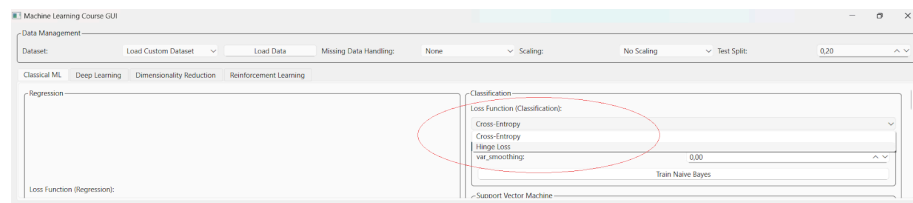
Regression Loss Functions:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Huber Loss



Classification Loss Functions:

- Cross-Entropy
- Hinge Loss



These loss functions are dynamically selected based on the user's input and integrated into the model training logic. The feature allows users to experiment with different loss functions, optimizing the model's performance.

2. Support Vector Machine (SVM) Implementation

SVM was implemented for both classification and regression tasks. The enhancements include:

Key Features of SVM Implementation:

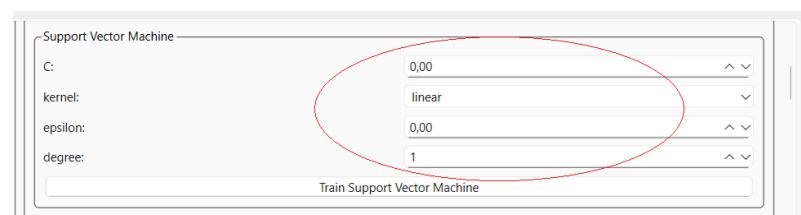
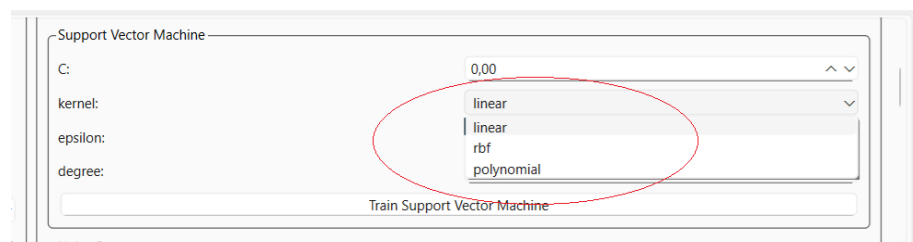
Kernel Selection:

- Linear
- Radial Basis Function (RBF)
- Polynomial

Hyperparameter Tuning:

SVM now supports tuning two key hyperparameters:

- C (Regularization) to control overfitting
- Epsilon for SVR (Support Vector Regression)

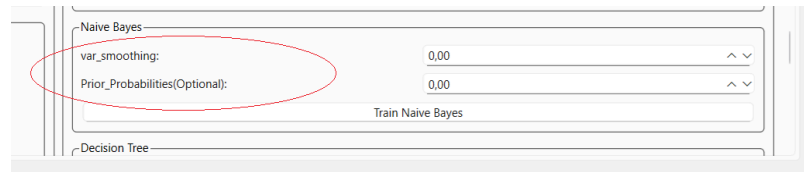


These selections allow the user to tailor the SVM model to different datasets and requirements. Users can experiment with different kernel types and hyperparameter values to improve the model's performance.

3. Gaussian Naive Bayes (GaussianNB)

The GaussianNB classifier was added, allowing users to adjust the `var_smoothing` parameter and select uniform or custom prior probabilities. This helps fine-tune the model and test different priors.

- **var_smoothing:** Prevents issues with zero variance.
- **Prior Probabilities:** Users can input custom prior probabilities (e.g., [0.3, 0.7]).



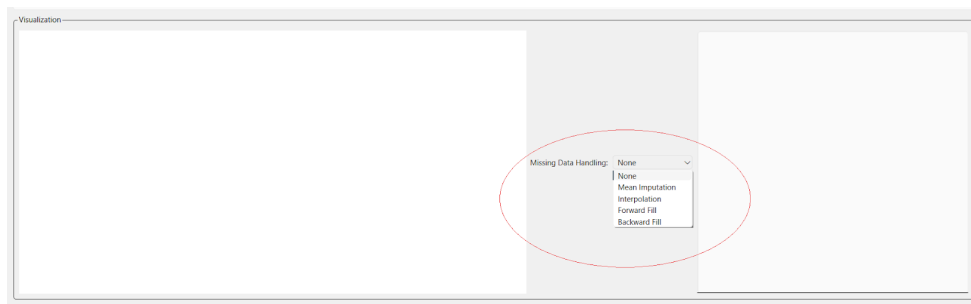
4. Missing Data Handling Methods

The GUI now supports handling missing data using scikit-learn's SimpleImputer. Users can select from the following options:

Mean Imputation: Replaces missing values with the column mean.

Interpolation: Estimates missing values using interpolation (e.g., linear).

Forward/Backward Fill: Fills missing values using the previous or next available value.



The logic was integrated to allow users to split the data into training and testing sets and apply the selected missing data handling method before training the model.

5. Model Evaluation and Testing

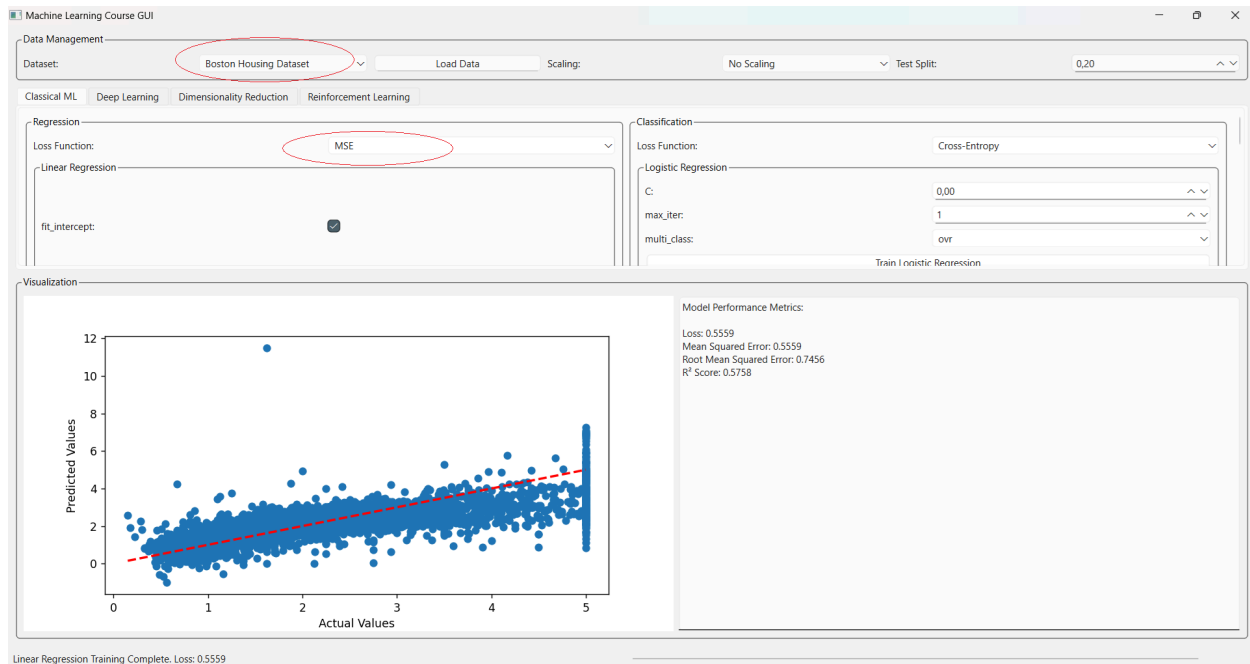
The implemented features were evaluated using different datasets for both regression and classification tasks.

➤ Regression Evaluation

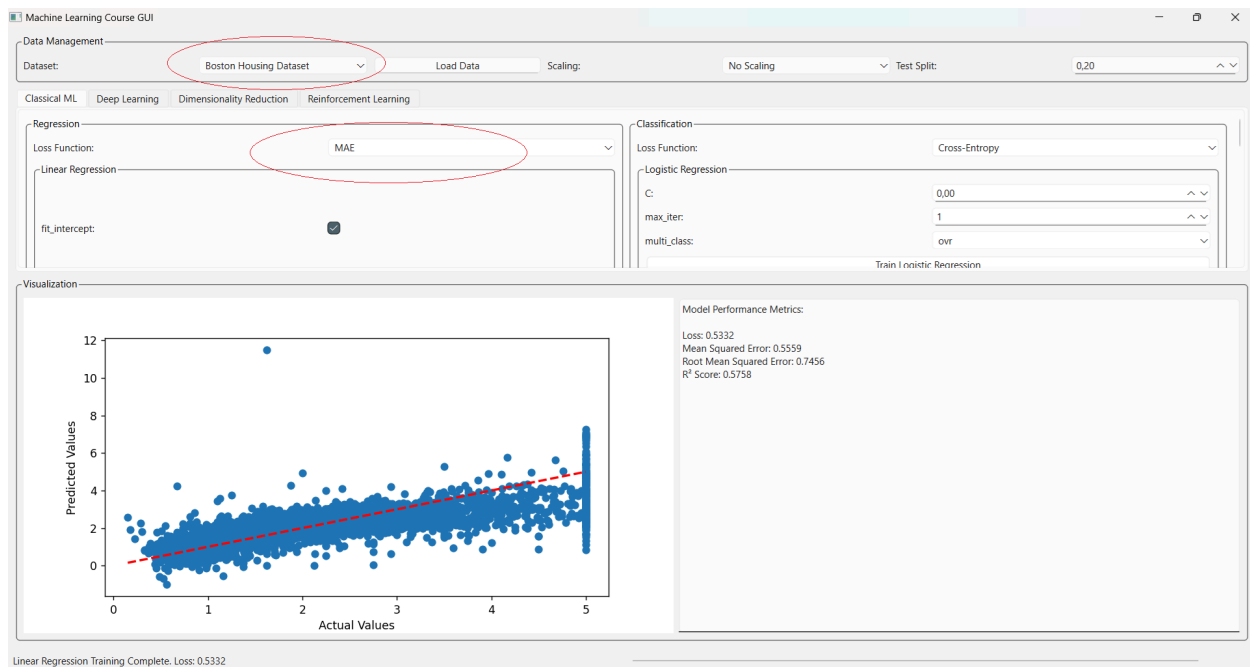
Dataset: Boston Housing Dataset

Model: Linear Regression

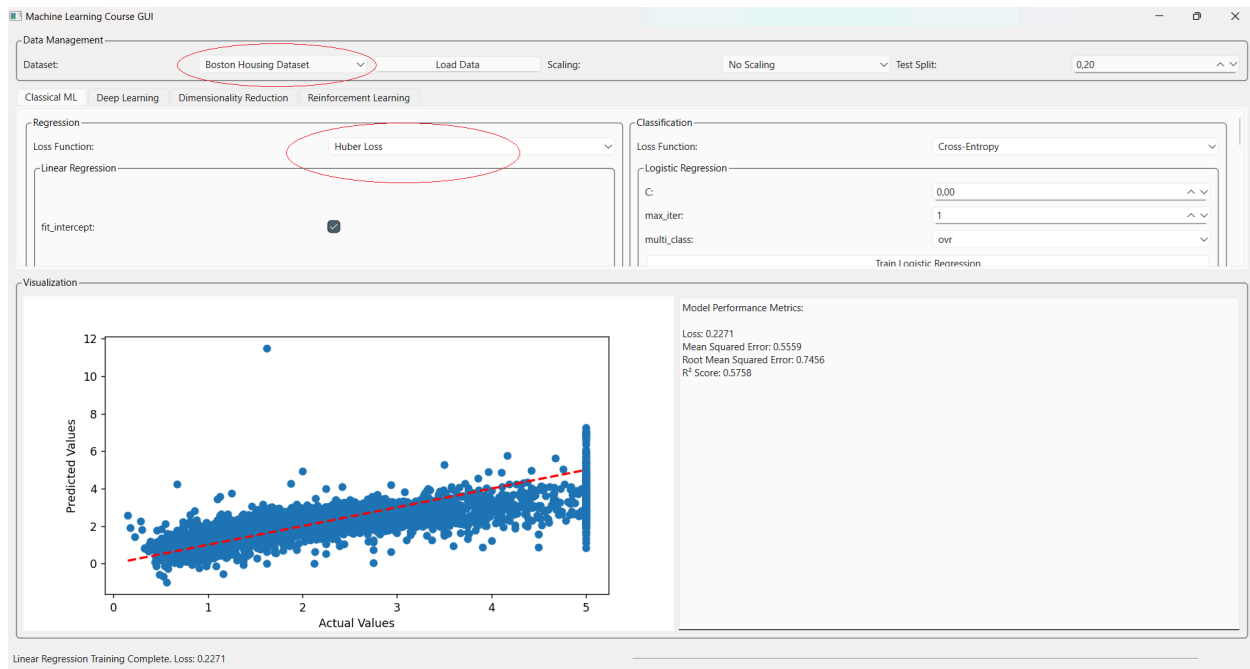
Loss Function: MSE



Loss Function: MAE



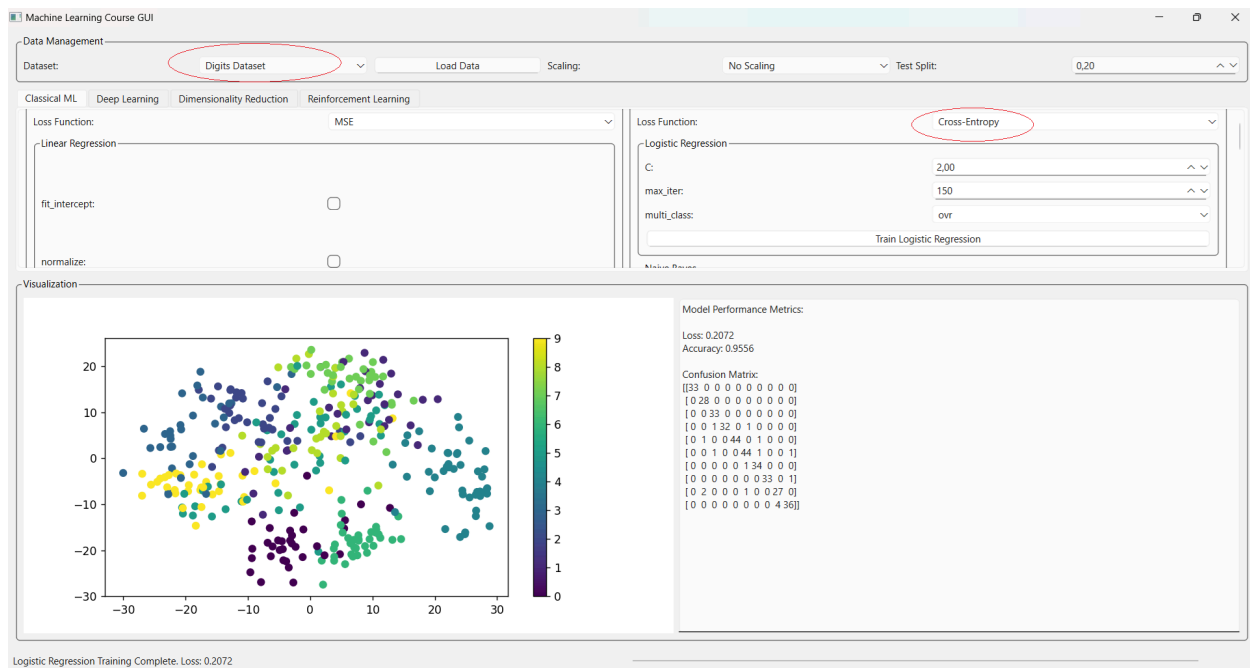
Loss Function: Huber Loss



➤ Classification Evaluation

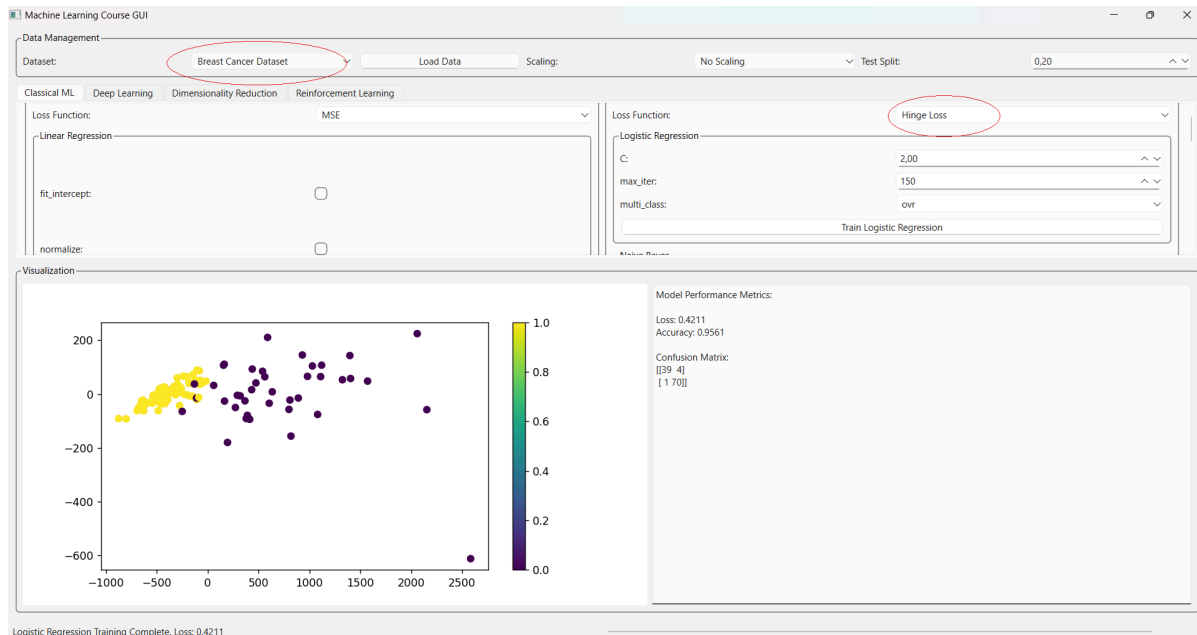
Dataset: Digits Dataset

Loss Function: Cross-Entropy



Dataset: Breast Cancer Dataset

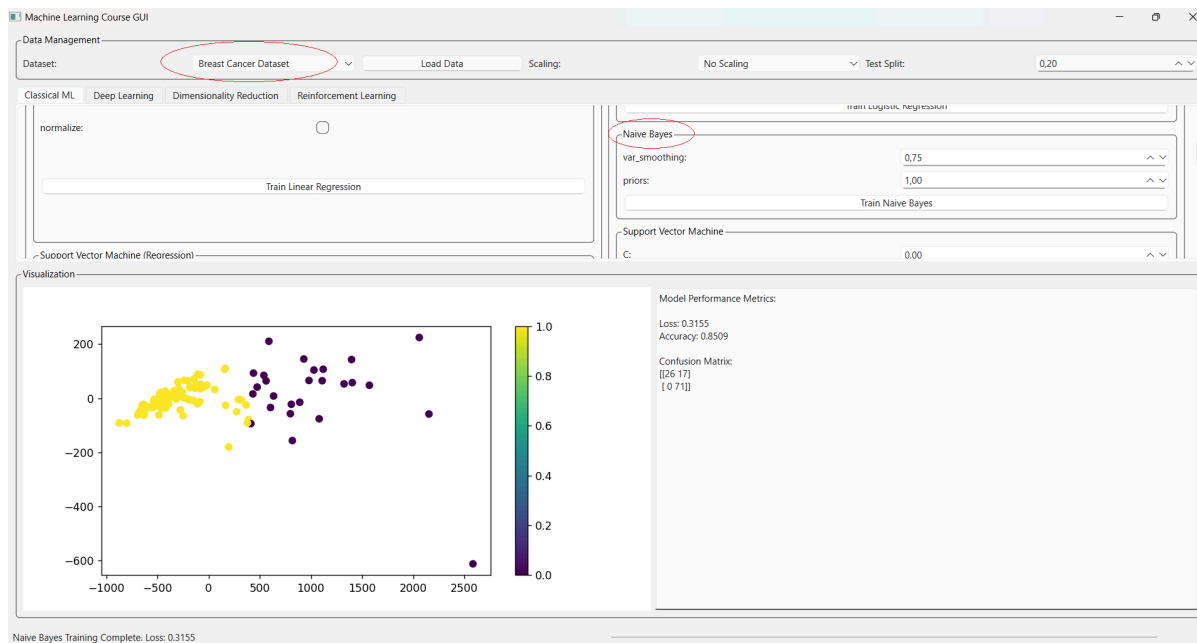
Loss Function: Hinge Loss



➤ Naive Bayes Evaluation

Dataset: Breast Cancer Dataset

Model: Gaussian Naive Bayes



6. Conclusion

This assignment improved the GUI with features like customizable **loss functions**, **SVM**, **Gaussian Naive Bayes**, and **missing data handling**, making it more versatile and user-friendly for machine learning tasks.